

RGB-Thermal cameras calibration based on Maximum Index Map

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ABSTRACT

Calibration across multiple sensors is crucial in computer vision tasks, such as image registration, 3D reconstruction, and target tracking. RGB-Thermal (RGB-T) cameras, widely used as a multi-sensor combination, traditionally depend on manual calibration methods to determine extrinsic parameters. However, this manual process is intricate and lacks the capability for online calibration during use. To overcome these challenges, an online calibration algorithm for RGB-T camera extrinsic parameters is proposed. The algorithm first addresses modal differences by matching feature point pairs using the Maximum Index Map (MIM) feature. These matched features are then used to calculate the extrinsic parameters through homography and epipolar constraints between images. Additionally, a convenient intrinsic calibration method is introduced, one that does not rely on extrinsic infrared light sources, thereby overcoming the limitations of standard calibration boards, which are often unsuitable for RGB-T cameras. Experimental results demonstrate that the proposed online extrinsic calibration algorithm significantly simplifies the calibration process compared to traditional methods, achieving accurate calibration with only a pair of images. The method achieves an average reprojection error of 0.677 pixels for RGB images and 0.635 pixels for thermal images, highlighting its precision. The method's effectiveness is further validated by the precision in RGB-T image registration.

1. Introduction

RGB-T cameras, capturing both visible and infrared spectrum information, offer complementary benefits in harsh and complex environments, thus enhancing system robustness against environmental disturbances [1]. This technology has seen wide application in fields such as autonomous driving [2], video surveillance [3], and robotics [4].

RGB-T cameras consist of independent RGB and thermal imaging systems. The misalignment of the optical axes between these two systems creates imaging disparities, which constitute the primary cause of RGB-T image misalignment. Image misalignment greatly decreases the accuracy of environmental perception methods that rely on RGB-T data [5]. Registering RGB-T images requires calibrating both the intrinsic and extrinsic parameters of the two imaging systems. The intrinsic parameters of a camera, determined during manufacturing, remain stable during normal usage. The extrinsic parameters of a multi-sensor system consist of translation and rotation vectors, which change with variations in the relative position between sensors. Parameter calibration based on the pinhole camera model can be completed using a stereo calibration process [6]. The calibration process for RGB cameras is already

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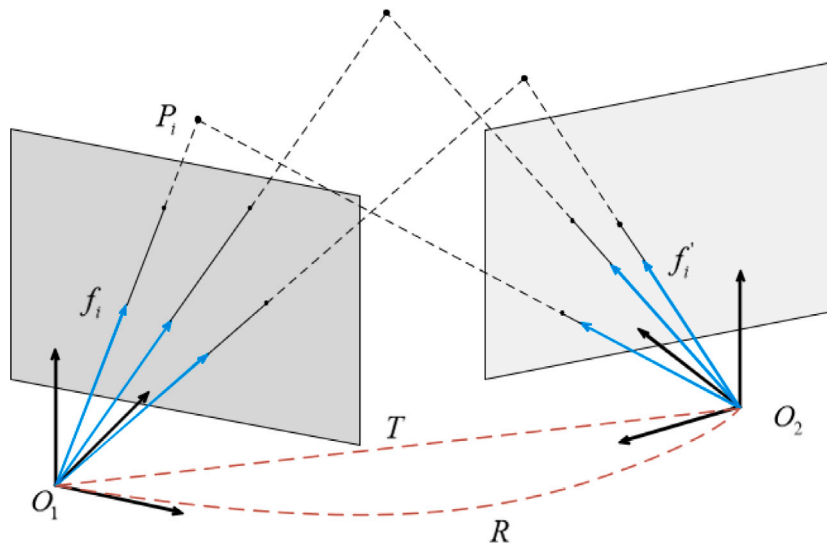


Fig. 1. Online extrinsic Calibration. This method estimates the relative translation vector T and relative rotation vector R of two cameras using a provided set of N pairs of unit direction vectors $\{f_i, f'_i\}$.

quite advanced, primarily utilizing calibration boards with checkerboard or other specific patterns [7–9]. However, thermal cameras are only sensitive to thermal radiation [10], making standard calibration boards unsuitable for their intrinsic calibration [11]. A popular method involves using a heated printed checkerboard [12], but this approach suffers from low accuracy due to the difficulty in maintaining temperature differences [13]. Yang et al. [14] and Zoetgnande et al. [15] arranged light bulbs on a checkerboard to achieve self-heating and maintain temperature differences; however, the high demands on the heat source made these methods difficult to implement. Mouats et al. [16] and Shivakumar et al. [17] used materials with different infrared emissivities to create infrared calibration boards that do not require a heat source, but their dependency on the usage environment limits their versatility. Afterward, Zhang et al. [18] proposed an optimized calibration method for dual-band cameras with ultra-wide angles based on a thermal radiation chessboard, improving mapping accuracy; Swamidoss et al. [19] developed a low-cost calibration board for thermal imaging cameras, achieving high contrast and reliable chessboard detection; Roshan et al. [20] employed ChArUco markers to achieve precise calibration of infrared cameras, resulting in higher-quality calibration. However, all of these methods are offline calibrations. In scenarios where extrinsic parameters frequently change, such as in vehicle-mounted applications, these offline calibration methods based on calibration boards require capturing dozens of images for re-calibration, leading to long maintenance cycles and high costs, ultimately affecting efficiency. Therefore, extrinsic parameter online calibration algorithms are particularly necessary.

Generally speaking, offline calibration tends to be more accurate than online calibration because it allows the use of more data and complex algorithms under controlled conditions. However, online calibration has the unique advantage of real-time adjustment and adaptability in dynamic environments. It is worth noting that most calibration methods still rely on offline calibration due to its superior accuracy and stability. While the proposed online calibration method may still lag behind the current advanced offline methods in terms of accuracy, it does offer significant improvements in online calibration. As shown in Fig. 1, online extrinsic calibration is achieved by calculating rotation and translation vectors from N pairs of feature points extracted from images captured by two calibrated sensors. This algorithm involves three main steps: extracting feature points between images of multiple sensors, calculating the 3D positions of these feature points, and estimating the extrinsic matrix between them [21,22]. However, for RGB-T multi-sensor combinations, a significant issue arises that cannot be overlooked: Traditional feature matching methods like Scale-Invariant Feature Transform (SIFT) [23], Speed Up Robust Features (SURF) [24], Oriented FAST and Rotated BRIEF (ORB) [25], etc., often fail to produce satisfactory results due to factors such as radiation distortion [26]. Mharolkar et al. [27] utilized the powerful feature extraction capabilities of neural networks to achieve online extrinsic calibration of RGB-T cameras using a pair of RGB-T images and point cloud data. Nakagawa et al. [28] achieved online extrinsic calibration of RGB-T cameras by acquiring 3D information using RGB-D cameras. However, both of these approaches do not solely rely on RGB-T images for extrinsic parameter computation; they leverage additional sensors to obtain depth information in advance, thereby reducing the complexity of feature computation. To our knowledge, there has been little research to date focusing solely on using RGB-T bimodal data for RGB-T cameras online extrinsic calibration.

In general, the traditional calibration board for RGB-T cameras is complex and inconvenient, highlighting the need for an intrinsic calibration method that does not rely on external infrared light sources. Furthermore, conventional extrinsic calibration processes are cumbersome and not suitable for online calibration, making them inadequate for dynamic applications. The significant differences between RGB and thermal imaging present additional challenges for cross-modal feature matching, which must be effectively addressed. To address these limitations, this paper proposes a new calibration method for RGB-T camera parameters.

Initially, an intrinsic calibration method for thermal cameras is presented, which does not require an infrared light source. By exploiting the radiation differences between aluminum and acrylic in the infrared spectrum, a calibration board is created that exhibits high contrast for thermal imaging, facilitating simultaneous intrinsic calibration of RGB and thermal cameras. Expanding on the completion of camera intrinsic calibration, an online calibration algorithm is proposed for the extrinsic parameters of RGB-T cameras. The algorithm establishes features through phase consistency and utilizes the Maximum Index Map (MIM) [26] for feature matching, effectively overcoming the challenges of multimodal feature matching. It then computes the RGB-T cameras extrinsic parameters using homography relationships and epipolar constraints. This algorithm does not rely on chessboard during the extrinsic calibration process, enabling precise online calibration of RGB-T cameras extrinsic parameters solely through real-time images captured by the sensors. This research holds significant importance for applications in fields of computer and electrical engineering, including autonomous driving, surveillance, power inspection, robotics, and more.

In summary, the contributions of this paper are as follows:

(1) Development of a novel intrinsic calibration method: A novel intrinsic calibration method for thermal cameras is proposed, eliminating the need for external infrared light sources. By exploiting the infrared radiation differences between aluminum and acrylic, a high-contrast calibration board is created, enabling the simultaneous calibration of both RGB and thermal cameras with just a single heating process.

(2) Introduction of an online extrinsic calibration algorithm: An online extrinsic calibration algorithm for RGB-T cameras is presented, which uses phase consistency for feature detection and applies Maximum Index Map (MIM) for cross-modal feature matching, effectively addressing the challenges of multimodal feature matching in calibration tasks.

2. Related works

2.1. Intrinsic calibration of RGB-T cameras

Intrinsic parameters of RGB-T cameras serve as a crucial basis for subsequent image rectification and registration. The intrinsic calibration of RGB cameras is relatively straightforward, commonly achieved by using a checkerboard as a calibration target and employing Zhang's calibration method [6] to obtain high-precision results. Due to the characteristics of sensors, the thermal camera exclusively captures images within the infrared spectrum, necessitating special calibration boards for intrinsic parameter calibration. Depending on whether a heat source needs to be provided, thermal camera calibration boards can be categorized into two types: active and passive.

Active calibration methods maintain contrast in thermal imaging by continuously transferring heat to the calibration board. Yang et al. [14] introduce a self-heating calibration board with 25 light bulbs in a 5×5 matrix to regulate its temperature. Zoetgnande et al. [15] design a circuit board with 36 light bulbs and developed an algorithm for subpixel corner estimation to identify thermal signals. While active calibration methods enhance thermal imaging contrast, their high demand for a heat source limits ease of use.

Passive calibration methods do not require artificial heating of the calibration board. Mouats et al. [16] proposed an effective passive calibration solution by applying black squares on the highly polished surface of an aluminum plate to create the calibration board. Under the influence of radiative cooling, temperature differences occur between the black and white regions of the calibration board, creating high contrast between the aluminum plate and black squares. Shivakumar et al. [17] attached a rough aluminum sheet to a black background board to create a calibration board. By using the human body as an infrared light source, sharp corners were generated in both RGB and thermal images through diffuse reflection. This approach requires the emitting area of the infrared light source to match the size of the checkerboard. Passive calibration methods eliminate the reliance on an extrinsic heat source, but they still depend on conditions such as a cold sky or an additional infrared light source during usage, making the calibration conditions demanding.

Addressing the aforementioned issues, a novel active intrinsic calibration method is introduced. As an active calibration approach, our method utilizes the difference in infrared radiation between aluminum and acrylic. Calibration is accomplished through a single heating process, mitigating the complexities and inconveniences linked to traditional active calibration methods. Furthermore, it guarantees robustness across various environmental conditions and eliminates the reliance on external infrared light sources.

2.2. Extrinsic calibration of RGB-T cameras

Extrinsic calibration across multiple cameras can be categorized into two types: offline calibration relying on calibration boards and online calibration based on feature matching [29]. Offline calibration has a relatively simple theory and wide applicability. However, the calibration process is cumbersome, and it cannot be performed online during usage. In scenarios where extrinsic parameters change, the system needs to be halted for recalibration. On the other hand, online calibration methods are suitable for scenarios where extrinsic parameters are unstable or need to be updated online, such as the initialization of stereo SLAM [28].

Traditional extrinsic calibration for RGB-T cameras mostly relies on offline calibration algorithms based on calibration boards [16, 17]. These methods involve detecting the corners of calibration boards in the imaging of both RGB and thermal cameras and subsequently performing stereo calibration to obtain the extrinsic parameters of the RGB-T cameras. However, such methods are not suitable for scenarios where the extrinsic parameters are subject to changes or online adjustments. In these cases, online calibration is a preferable solution. Unlike the calibration of unimodal stereo cameras, RGB-T cameras face significant challenges in cross-modal feature matching due to radiation distortion between RGB and thermal images.

Recently, the research on online extrinsic calibration for RGB-T cameras has been gradually evolving. While online calibration has become a growing topic in the field of multi-sensor fusion, there are relatively few methods specifically tailored for RGB-T cameras. For example, some works have explored calibration techniques that combine RGB and thermal cameras with other sensors, such as LiDAR, to perform extrinsic calibration in dynamic environments [27,28]. However, these approaches often rely on auxiliary hardware and are not exclusively focused on the RGB-T camera pair.

Traditional feature matching methods for online calibration, such as SIFT [23], SURF [24], and ORB [25], have been widely used in calibration tasks. These methods detect and match keypoints across images, and have been proven effective in scenarios where the imaging conditions are similar. However, due to the significant differences between RGB and thermal images—such as variations in texture, color, and intensity—traditional methods struggle to establish reliable feature correspondences across modalities. Moreover, these methods are sensitive to noise and illumination changes, further limiting their robustness for RGB-T camera calibration.

Another research direction is using learning-based methods for online extrinsic calibration. In recent years, deep learning-based feature matching methods, such as LightGlue [30], SuperGlue [31], and LoFTR [32], have shown promising results in traditional image matching tasks. These methods excel in complex scenarios, especially for matching within the same modality. However, these neural network-based approaches are computationally expensive, require high-performance hardware, and rely on large amounts of training data. Additionally, whether these methods are directly applicable to RGB-T cameras remains unknown due to the substantial differences between RGB and thermal images.

In recent years, cross-modal image registration has developed rapidly. Sun et al. [33] proposed an RGB-T image registration method that combines global structure and local mutual information. This method uses mutual information to provide an initial estimate for feature matching, enabling successful image registration. However, this method is highly time-consuming. Jiang et al. [34] introduced a novel feature point orientation extraction method called Contour Angle Orientation (CAO), which addresses the challenges of automatic registration between infrared and visible images for power equipment. However, the registration accuracy decreases when distinct contour features are absent. Ye et al. [35] proposed a multi-modal remote sensing image registration method based on image structural similarity, using a feature descriptor known as the Directional Phase Consistency Histogram (HOPC). However, this method is also quite time-consuming and unsuitable for real-time processing tasks. Overall, traditional online calibration methods fail to address the issue of radiometric distortions, while most methods capable of cross-modal registration are not suitable for fast registration tasks.

Given these challenges, as well as the lack of standardized datasets and datums specific to RGB-T camera calibration, online calibration of RGB-T cameras is still less prevalent compared to unimodal or other multimodal systems (e.g., RGB-D or RGB-LiDAR). Currently, research on online extrinsic calibration for RGB-T cameras remains in its early stages, with many unresolved issues. To address the need for online extrinsic parameter updates in applications such as in-vehicle scenarios, an online calibration method using MIM is proposed. This method not only meets the needs of online calibration but also effectively addresses the cross-modal problem. It achieves cross-modal feature matching through MIM features, facilitating the online calibration of extrinsic parameters for RGB-T cameras. This approach is more efficient and adaptable, offering a viable solution for real-time calibration updates in dynamic environments.

3. Method

This paper presents an RGB-T camera calibration method that uses a specially designed infrared calibration board to perform intrinsic calibration of the RGB-T cameras. Based on this, the extrinsic parameters of the RGB-T cameras can be calibrated online directly from the images captured by the cameras. The method architecture is illustrated in Fig. 2. The method begins with intrinsic calibration, where a custom calibration board is used to estimate the camera's internal parameters. Once intrinsic calibration is complete, these parameters are utilized in the feature matching process between the RGB and thermal images. At this stage, the Maximum Index Map (MIM) method is applied for feature description and matching across modalities. After identifying the matched feature points, the extrinsic calibration is conducted, leveraging these matched points to compute the transformation between the RGB and thermal cameras. This overall workflow ensures robust and accurate cross-modal calibration.

3.1. Intrinsic calibration of RGB-T cameras

Enhancing the radiation difference between the target and background is crucial in thermal imaging to improve image contrast. Regarding checkerboard images, it is necessary to maximize the radiation difference between the black and white squares of the checkerboard. According to thermodynamic principles, the total radiation detected by the detector from opaque objects comprises both the radiation reflected by the object and the energy radiated outward from the object itself, as follows:

$$W = \epsilon W_b + (1 - \epsilon)W_r \quad (1)$$

Where ϵ is the emissivity of the object; W_b is the radiation value of a blackbody at the same temperature as the object; and W_r is the energy value of environmental reflection. Amplifying the emissivity difference between the black and white squares is essential for increasing contrast in the checkerboard images. For the black squares, matte black acrylic with an emissivity of approximately 0.93 [36] is employed, and for the white squares, polished aluminum with an emissivity of approximately 0.05 [37] is used. These materials demonstrate significant differences in both RGB and thermal imaging.

Additionally, aluminum's high specific heat capacity and low thermal conductivity [38] lead to a slow rate of heat dissipation in the air. Consequently, the temperature of aluminum does not drop immediately. During usage, applying heat from a heat source

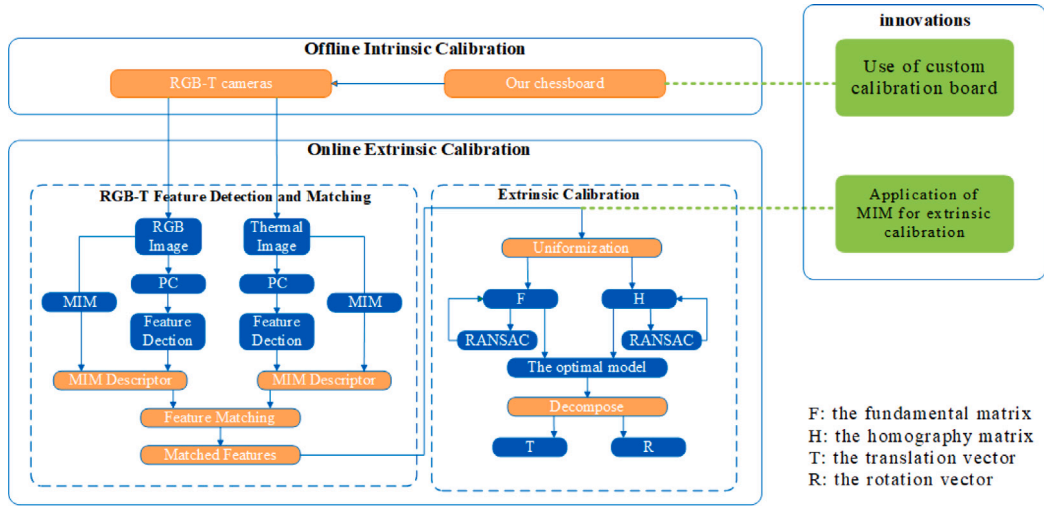


Fig. 2. The proposed method architecture. It consists of three main parts: Intrinsic Calibration, RGB-T feature detection and matching, and extrinsic calibration. Text boxes with yellow backgrounds represent the overall workflow, while those with blue backgrounds show detailed steps.



Fig. 3. Chessboard corner detection results. (a) RGB. (b) Thermal.

such as a human body or a heat gun to the calibration board once creates long-lasting high contrast. This ensures that the calibration board maintains high contrast for an extended period. Experimental results indicate that the calibration board maintains contrast for about 5 min, sufficient for completing one calibration. As illustrated in Fig. 3, the designed calibration board can simultaneously produce clear and distinguishable corner point images under both RGB and thermal cameras. After obtaining sharp corner point images, Zhang's calibration method [6] is utilized to determine the intrinsic parameters of the RGB and thermal cameras (see Fig. 3).

3.2. RGB-T feature detection and matching

Image decomposition through Fourier transform yields amplitude and phase components. The phase information exhibits high invariance to illumination and scale variations in the image [39]. Log-Gabor filters are employed to extract phase information from images. The log-Gabor filter in the log-polar coordinates of the Fourier domain is defined as follows:

$$G_{(s,o)} = \exp\left(-\frac{1}{2}\left(\frac{\rho - \rho_s}{\sigma_\rho}\right)^2\right) \exp\left(-\frac{1}{2}\left(\frac{\theta - \theta_{(s,o)}}{\sigma_\theta}\right)^2\right) \quad (2)$$

Where (ρ, θ) represents the coordinates under $\log_2$ scale; s and o denote the scale and orientation of the filter; $(\rho_s, \theta_{(s,o)})$ denotes the center frequency of the filter; $(\sigma_\rho, \sigma_\theta)$ represents the bandwidth of (ρ, θ) . After convolving the image with the log-Gabor filter, both amplitude and phase components are obtained. Feature detection is then performed by calculating phase consistency (PC), which is computed using the equation as follows:

$$PC(x, y) = \frac{\sum_s \sum_o \omega_o(x, y) [A_{so}(x, y) \Delta \Phi_{so}(x, y) - T]}{\sum_s \sum_o A_{so}(x, y) + \xi} \quad (3)$$

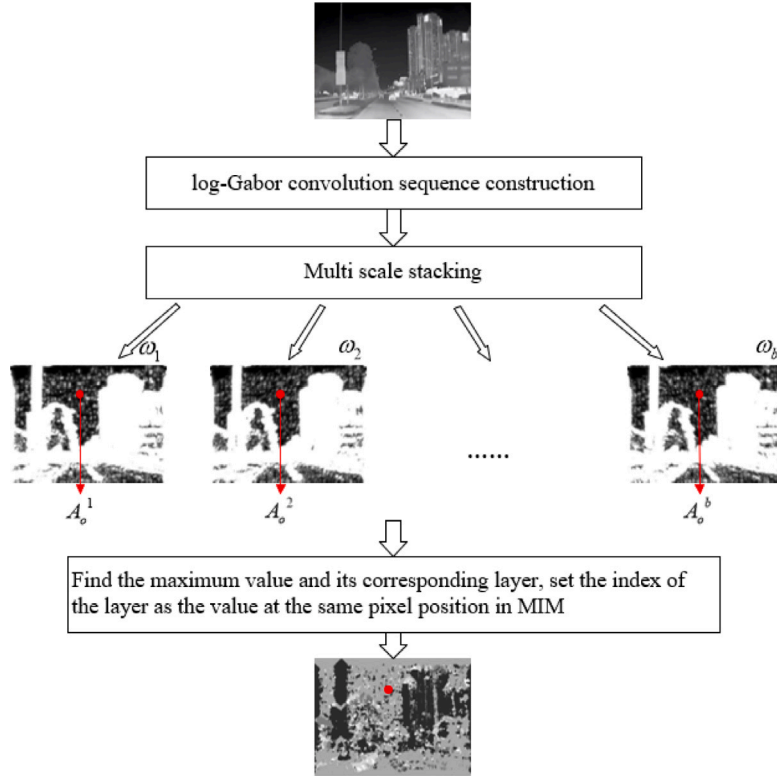


Fig. 4. The construction of a MIM.

Where (x, y) represents the pixel coordinates of the image; ξ is a small value introduced to avoid division by zero; T is the noise intensity; $A_{so}(x, y)$ represents the intensity information of the image; $\Delta\Phi_{so}(x, y)$ is the phase deviation function; $\omega_o(x, y)$ is the weight function. For specific calculation details, please refer to [40]. Following the computation of phase consistency, the covariance matrix is computed, and its minimum and maximum moments are calculated to generate a highly localized operator. This operator is then used in a contrast-invariant manner to identify edges and corners. The construction of the covariance data is as follows:

$$Covx(\theta_o) = PC(\theta_o) \cos(\theta_o) \quad (4)$$

$$Covy(\theta_o) = PC(\theta_o) \sin(\theta_o) \quad (5)$$

$$a = \sum_o Covx(\theta_o)^2 \quad (6)$$

$$b = \sum_o Covx(\theta_o)Covy(\theta_o) \quad (7)$$

$$c = \sum_o Covy(\theta_o)^2 \quad (8)$$

Where $PC(\theta_o)$ is the PC mapping for the angle θ_o . The computation of the minimum and maximum moments is given by:

$$M = \frac{1}{2}(c + a + \sqrt{b^2 + (a - c)^2}) \quad (9)$$

$$m = \frac{1}{2}(c + a - \sqrt{b^2 + (a - c)^2}) \quad (10)$$

The maximum moment M functions as the edge map of the image, marked as an edge if it surpasses a certain threshold. The minimum moment m is employed for corner detection in the image, flagged as a corner if its value exceeds a certain threshold.

The algorithm proposed in this paper addresses the challenge of nonlinear radiation distortions (NRD) in RGB-T cross-modal imaging by constructing the Maximum Index Map (MIM) for feature description and matching. NRD typically causes inconsistencies in the radiometric information between RGB and thermal images, which makes it difficult for traditional feature matching methods to perform reliably. In contrast, MIM mitigates the impact of NRD by focusing on extracting the most prominent directional features

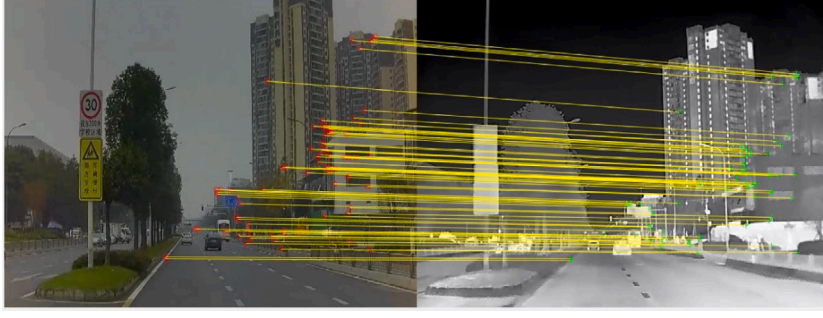


Fig. 5. Feature matching based on MIM. The red circles and the green crosshairs in the figure indicate the feature points on the RGB image and the thermal image, respectively; the yellow lines indicate the correct matches.

at multiple scales and orientations. This approach enables MIM to capture structural features that are more consistent and robust against radiometric distortions, providing a reliable solution for feature matching in RGB-T cross-modal images.

By selecting the maximum response across different scales and orientations, MIM identifies the most significant structural features in the image. This ensures that the most relevant features are preserved, making MIM less sensitive to NRD compared to traditional methods that might rely on less robust features. Fig. 4 illustrates the process of MIM construction. In this process, the original image is convolved with log-Gabor filters applied at a scales and b orientations, resulting in $a \times b$ amplitude images. The elements labeled as $\omega_1, \omega_2, \dots, \omega_b$ in the figure represent the different orientation angles used during the log-Gabor filtering process. These angles help capture features in various directions, ensuring that important directional information is retained at different scales.

Subsequently, the images from the same orientation but different scales are summed to create a new feature sequence consisting of b overlaid images. The grayscale values at the same pixel coordinates across these b images are compared, and the index corresponding to the maximum grayscale value is recorded as the grayscale value of the MIM image. This process ensures that each pixel retains the most prominent directional feature, which is less affected by radiometric distortions. The construction of MIM is described by the following equation:

$$A_o(x, y) = \sum_{i=1}^p A_{so}(x, y) \quad (11)$$

$$MIM(x_j, y_j) = \operatorname{argmax}\{\{A_o^i(x_j, y_j)\}_1^q\} \quad (12)$$

Where $A_{so}(x, y)$ represents the filtered image; $A_o(x, y)$ is the image after scale summation; i denotes different channels of the feature sequence; (x_j, y_j) represents any point in $A_o(x, y)$; $MIM(x, y)$ is the constructed Maximum Index Map. After obtaining the Maximum Index Map (MIM), feature points are extracted by identifying the locations of the maximum values within the MIM. These feature points are typically found in the most prominent regions of the image, such as edges, corners, or areas with rich textures. The distribution of these feature points covers the significant regions of the image, effectively capturing local image characteristics that are essential for accurate feature matching. To describe these feature points in detail, distribution histograms are constructed for each feature point. A window region is defined around each feature point, and the pixel value distribution within this region is calculated. This distribution is represented by a histogram, where each bin corresponds to the frequency or count of pixel values within the window. These histograms serve as descriptors for the feature points, capturing and expressing local image features, which are critical for subsequent image matching and analysis. Feature matching is then performed by comparing the feature vectors using Euclidean distance as the metric. Fig. 5 illustrates some successfully matched feature point pairs between the RGB and thermal images. This matching process ensures that even in the presence of NRD, MIM can identify and match significant features reliably, providing a robust solution for RGB-T cross-modal imaging applications.

3.3. Extrinsic calibration for RGB-T cameras

The 8-point algorithm [41] is used to estimate the extrinsic parameters. Considering that coplanar feature points can lead to the failure of the 8-point algorithm, a strategy is employed to simultaneously compute the homography matrix and the fundamental matrix between the RGB and thermal images. The specific calculation steps are as follows:

(1) Compute the two models in parallel:

Uniformize feature point pairs using a quadtree algorithm. Subsequently, feed the uniformly distributed feature point pairs into two sets of RANSAC loops, calculating the inliers for both the homography matrix and the fundamental matrix. By leveraging the homographic relationship between two planes and the epipolar constraint, the inliers are input to compute the homography matrix and the fundamental matrix as follows:

$$(X_{Thermal})^T F(X_{RGB}) = 0 \quad HX_{RGB} = X_{Thermal} \quad (13)$$

Where X_{RGB} is the homogeneous coordinates of the feature points in the RGB image; $X_{Thermal}$ is the homogeneous coordinates of the feature points in the thermal image; F is the fundamental matrix; and H is the homography matrix.

(2) Select the optimal model:

The algorithm in this paper adopts the scoring strategy from ORB-SLAM [42] to evaluate the fundamental matrix and the homography matrix. The scores S_F and S_H for the fundamental matrix and homography matrix, respectively, are obtained by calculating the bidirectional reprojection errors [43]. Then, the two models are compared using the following equation:

$$P_H = \frac{S_H}{S_H + S_F} \quad (14)$$

When $P_H \geq 0.45$, the homography matrix is chosen as the optimal model to avoid degeneration of the 8-point algorithm due to low disparities and coplanar feature points. In other cases, the fundamental matrix is chosen.

(3) Recover the extrinsic parameters from the optimal model:

After obtaining the optimal model, the translation vector T and rotation vector R are recovered from it. If the optimal model obtained in the second step is the homography matrix, a transformation is applied to eliminate the impact of inconsistent camera intrinsic parameters between the RGB and thermal cameras, as follows:

$$H_N = (K_{Thermal})^{-1} H (K_{RGB}) \quad (15)$$

Where $K_{Thermal}$ is the intrinsic matrix of the thermal camera, K_{RGB} is the intrinsic matrix of the RGB camera. The singular value decomposition (SVD) of H_N is performed to obtain the translation and rotation vectors. If the optimal model obtained is the fundamental matrix, the intrinsic matrices of both cameras are used to obtain the essential matrix E , as follows:

$$E = (K_{Thermal})^T F (K_{RGB}) \quad (16)$$

Performing SVD decomposition on the essential matrix yields translation and rotation vectors. It is essential to note that the decomposition results of the homography matrix and the essential matrix have multiple solutions: the homography matrix has 8 sets of solutions, and the essential matrix has 4 sets of solutions. To determine the correct solution, this algorithm utilizes triangulation to solve for the three-dimensional coordinates of feature points in the normalized camera coordinate system. The combination with positive feature point depth is then selected as the correct result.

3.4. Evaluation metrics for calibration results

The reprojection errors (RPE) are calculated to verify the accuracy of the RGB-T camera calibration results. First, the 3D coordinates of feature points are reconstructed using the calibrated camera parameters. The 3D points are then projected back onto the RGB image plane and thermal image plane, obtaining the reprojection pixel coordinates [44]. To better analyze the errors, the Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Standard Deviation (SD) of RPE are computed. Their definitions are as follows:

$$R_{PE} = \sqrt{(x(i, j) - \hat{x}(i, j))^2} \quad (17)$$

$$R_{MSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (R_{PE})^2} \quad (18)$$

$$M_{AE} = \frac{1}{n} \sum_{i=1}^n |R_{PE}| \quad (19)$$

$$S_D = \sqrt{\frac{1}{n} \sum_{i=1}^n (R_{PE} - \overline{R_{PE}})^2} \quad (20)$$

Where

$$\hat{x} = \frac{1}{s} K P X \quad (21)$$

$$\overline{R_{PE}} = \frac{1}{n} \sum_{i=1}^n R_{PE} \quad (22)$$

Here, x represents the 2D coordinates of the feature point; $\hat{x}$ represents the 2D coordinates of the feature point after reprojection; X represents the 3D coordinates of the feature point; P is the projection matrix; K is the camera intrinsic matrix; s is the depth of the feature point.

4. Experiment

4.1. Experimental setup

As shown in Fig. 6, the thermal camera used is the Iray pilot640 with a pixel pitch of 12um, a focal length of 13 mm, and a resolution of 640×512 . The RGB camera is the SG1-OX01F10C-GMSL-H30 A with a pixel pitch of 3um, a focal length of 6 mm, and a resolution of 1280×720 . By performing scaling and cropping operations, the resolutions of both cameras were adjusted to 640×480 .



Fig. 6. RGB-T camera setup. The RGB camera and the thermal camera are mounted on a sliding rail, with the relative position adjusted to ensure adequate overlap in their fields of view. Left: Thermal camera; Right: RGB camera.

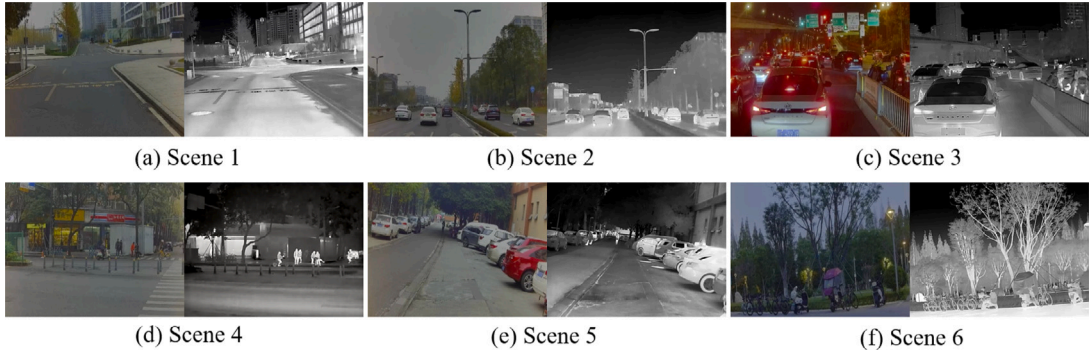


Fig. 7. Experimental data.

4.2. Dataset

Experimental data were collected in six different scenes, as shown in Fig. 7, to validate the effectiveness of the proposed algorithm. The dataset exhibits significant variation in image feature density, specifically with some images having sparse feature points (< 150) and others having dense feature points (> 450). This disparity allows us to evaluate the performance of our algorithm under different feature density conditions.

4.3. Experimental procedures

(1) RGB-T Cameras Calibration: initially, the intrinsic parameters of the RGB-T cameras are calibrated using 35 sets of checkerboard images with clear imaging, leading to accurate corner detection. Zhang's calibration method [6] is applied to derive the intrinsic matrices for both the RGB and thermal cameras. After obtaining the intrinsic parameters of the cameras, the proposed algorithm is used for online calibration of the extrinsic parameters of the RGB-T cameras in six different scenes, followed by analysis of reprojection errors.

(2) RGB-T Image Registration: The camera parameters of the RGB-T cameras are applied to achieve image registration, and the accuracy of the proposed method is further validated through the registration results. To register the images, the RGB image is first mapped to the normalized camera coordinate system, using the average depth of feature points as the scene depth. Then, a projection transformation is performed using the extrinsic parameters, aligning the RGB image with the thermal image. The registration calculation equation is as follows:

$$X' = \frac{1}{s_1}(s_2 K_{Thermal} R K_{RGB}^{-1} X + K_{Thermal} T) \quad (23)$$

Where X represents the homogeneous coordinates of the RGB image; and X' represents the homogeneous coordinates after registration; s_1 and s_2 are the average depths of feature points in the RGB and thermal camera coordinate systems, respectively. The registration results are shown in Fig. 5, it can be observed that the registered RGB image is well-aligned with the thermal image, affirming the reliability and accuracy of the parameters calibrated by the algorithm proposed in this paper.

Table 1

Results of intrinsic calibration. (f_x, f_y) are the horizontal and vertical focal lengths; (u, v) are the coordinates of the camera principal point. All values are in pixels.

		f_x	f_y	u	v
RGB	Theoretical value	1000.00	1333.33	320.00	240.00
	Calibration value	981.17	1307.98	315.97	248.04
Thermal	Theoretical value	1083.33	1083.33	320.00	256.00
	Calibration value	1092.71	1084.56	315.74	270.34

Table 2

Results of extrinsic calibration. T represents the translation vector, which becomes dimensionless after normalization; R represents the rotation vector, measured in degrees.

Parameters	$T = [t_x, t_y, t_z]$	$R = [r_x, r_y, r_z]$
Scene1	[-0.9872 0.0176 0.1587]	[-0.0957 -0.0365 -0.0621]
Scene2	[-0.9881 0.0129 0.1528]	[-0.0612 0.1317 0.0651]
Scene3	[-0.9871 0.0193 0.1586]	[-0.0367 -0.0879 0.0194]
Scene4	[-0.9884 0.0153 0.1513]	[0.0124 -0.0425 -0.0718]
Scene5	[-0.9870 0.0161 0.1600]	[-0.0916 -0.0981 -0.0409]
Scene6	[-0.9888 0.0177 0.1481]	[-0.1092 0.1013 -0.0616]

(3) Comparison with Existing Methods: The proposed algorithm is compared with five representative high-precision methods, using Mean Reprojection Error (MRE) as the unified evaluation metric. The formula for calculating MRE is as follows:

$$M_{RE} = \frac{1}{n} \sum_{i=1}^n |R_{PE}| \quad (24)$$

The selected methods include three based on active calibration boards [14,20,45] and one based on a passive calibration board [16].

5. Results and discussion

5.1. Calibration results for RGB-T cameras parameters

Table 1 shows the intrinsic parameters obtained from the calibration. Notably, due to scaling, the focal lengths in the horizontal and vertical directions differ for the RGB camera, while for the thermal camera, the principal point's position is not at the image center due to cropping. Table 2 shows the results of extrinsic parameters obtained in six different scenes. As the absolute scale of translation between camera coordinate systems cannot be determined, normalization is applied, fixing the magnitudes of the six translation vectors to 1.

5.2. RGB-T image registration

First, we compared the MIM algorithm with traditional methods such as SIFT [23], ORB [25], AKAZE [46], LSD [47], and SURF [24] in terms of feature point matching, as shown in Fig. 8. As can be seen from the figure, the proposed method significantly outperforms the other five algorithms, which is consistent with our expectations. Due to factors like radiometric distortions, traditional algorithms show poorer performance in feature matching. Generally, when the algorithm performs well, the matched feature points will form nearly parallel lines, indicating high matching accuracy. From the figure, it is clear that the performance of other methods is relatively poor, which is why we did not proceed with accuracy comparisons.

Fig. 9 shows the registration results. It can be observed that the registered RGB images align well with the thermal images. Qualitative analysis confirms that the extrinsic parameters calibrated by our algorithm are reliable and accurate.

5.3. Analysis of reprojection errors in different scenes

Fig. 10 shows the results of the RPE. The reprojection errors are concentrated in the range of ± 1 pixel, with the majority falling within ± 1.5 pixels. Some points exhibit horizontal errors exceeding the ± 1.5 pixel range. This outcome primarily due to the x-component dominating the translation between camera coordinate systems. Consequently, the epipolar lines are primarily oriented in the horizontal direction, with the epipolar constraints mainly influencing the vertical direction. For further analysis of the extrinsic calibration results, the RMSE, MAE, and SD of the RPE are calculated, as presented in Table 3. The error analysis results indicate that the proposed algorithm achieves accurate calibration of RGB-T camera parameters. To further explore the impact of different factors on calibration accuracy, the relationship between the number of feature points and RMSE across six scenes was compared. The results are shown in Fig. 11.

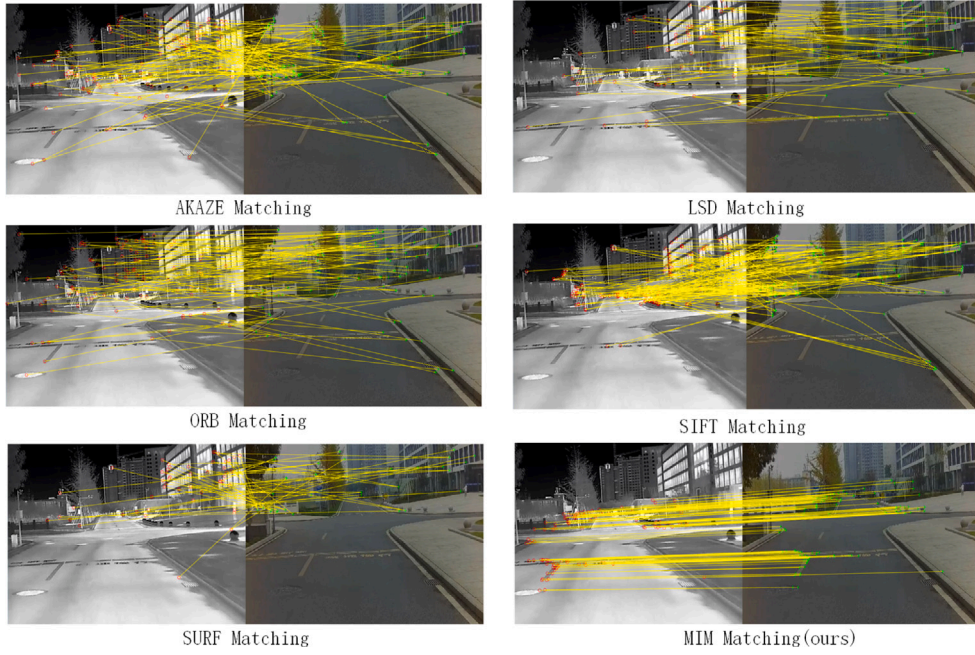


Fig. 8. Comparison of three different feature matching algorithms.



Fig. 9. RGB-T image registration results. For better comparison, appropriate cropping of the images has been performed.

Table 3

Quantitative comparison of the reprojection errors of the RGB camera and the thermal camera by using the RMSE, MAE, and SD. All values are in pixels.

	Scene1		Scene2		Scene3	
	RGB	Thermal	RGB	Thermal	RGB	Thermal
RMSE	1.160	1.179	1.172	1.015	0.804	0.782
MAE	1.023	0.986	1.137	0.982	0.722	0.698
SD	0.549	0.647	0.284	0.260	0.355	0.352
	Scene4		Scene5		Scene6	
	RGB	Thermal	RGB	Thermal	RGB	Thermal
RMSE	0.884	0.864	0.820	0.789	0.786	0.752
MAE	0.786	0.754	0.722	0.705	0.677	0.635
SD	0.405	0.422	0.367	0.354	0.398	0.404

(1) Impact of Scene Complexity:

Through comparisons of different scenes, an inverse relationship is observed between the number of feature points and the accuracy of the calibration algorithm. In complex scenes, such as scene 3 and scene 5 (with many vehicles) and scene 6 (with many bicycles), the higher number of feature points led to a lower RMSE. This is because complex scenes offer more

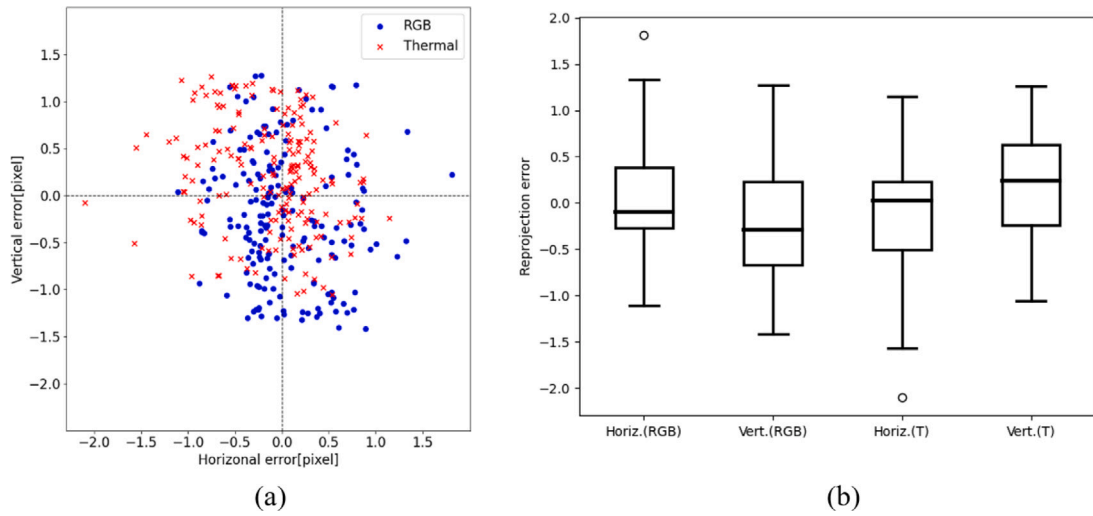


Fig. 10. The reprojection errors for the RGB camera and the thermal camera. (a) Distribution plot. (b) Statistical plot.

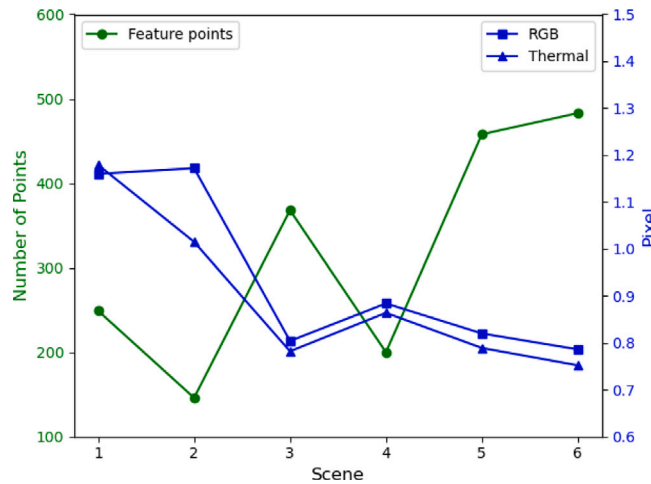


Fig. 11. Relationship between RMSE and number of feature points.

textures and geometric structures, providing richer feature points that enhance the robustness of cross-modal matching. In contrast, simpler scenes like scene 1, with fewer feature points, resulted in higher RMSE. This could be due to the increased likelihood of mismatches when fewer feature points are available, reducing the overall calibration accuracy. Hence, the density of feature points directly affects the precision of the calibration, explaining why the method performs significantly better in complex scenes.

(2) Impact of Lighting Conditions:

The effect of lighting on the calibration algorithm is not entirely linear. In scenes with insufficient lighting, such as scene 3 and scene 6, while RGB image feature detection may be hindered, thermal imaging remains stable, which helps maintain a lower RMSE overall. Additionally, scene 5, which had sufficient lighting, also showed low errors, indicating that the algorithm adapts well to different lighting conditions. This robustness is primarily due to the thermal imaging's insensitivity to lighting variations, ensuring high accuracy in cross-modal matching even when the lighting conditions are less than ideal. Even in scenes where RMSE is higher, the error remains within reasonable thresholds, further demonstrating the method's stability across various lighting conditions.

5.4. Comparison of reprojection errors across different calibration methods

Table 4 presents the comparison of the MRE metrics between our algorithm and other methods. For stereo systems, achieving an MRE of less than 2 pixels is generally considered adequate for 3D reconstruction [48]. Offline extrinsic calibration methods based

Table 4
Comparison of MRE metrics.

Method	Calibration Mode	RGB (pixel)↓	T (pixel)↓
Mouats et al. [16]	Offline (passive)	0.701	0.709
Roshan et al. [20]	Offline (active)	0.380	0.392
Yang et al. [14]	Offline (active)	0.586	0.572
Sun et al. [45]	Offline (active)	0.277	0.100
Ours	Online	0.677	0.635

on calibration boards can typically keep the MRE within 1 pixel, with active methods achieving higher precision due to their higher contrast calibration boards.

Our proposed algorithm falls under online extrinsic calibration and achieves an MAE of less than 1 pixel without relying on calibration boards, which is lower than the passive calibration methods listed in the table. Although the MAE of our algorithm is slightly higher than that of active calibration methods, it offers significant advantages in applications where external parameters are unstable or require real-time updates. Unlike active calibration methods, which often require more complex calibration processes and additional equipment, our approach simplifies the calibration process by eliminating the need for calibration boards. This makes it particularly advantageous in dynamic scenarios where real-time extrinsic calibration is required.

As shown in Table 4, while our algorithm achieves a reasonable MRE, there remains a gap compared to the top-performing offline methods that rely on calibration boards, such as those by Roshan et al. [20] and Sun et al. [45]. The key advantage of our method lies in its speed and ease of calibration without the need for additional calibration hardware. This makes it more suitable for applications in dynamic environments where external parameters need to be updated frequently or in scenarios where calibration equipment is not readily available. Despite the MRE gap, the simplicity and rapid deployment of our approach provide a strong solution for real-time calibration in such contexts.

6. Conclusion

The calibration of RGB-T cameras is the crucial prerequisite for accurately implementing environmental perception technology based on RGB-T multispectral imaging. In this paper, to address the complexities of fabrication and the inconvenience of traditional calibration targets, an efficient intrinsic calibration method is presented. This method enables accurate calibration for RGB-T cameras without the need for external infrared light sources. Furthermore, to tackle the cumbersome procedures and inability to calibrate online with traditional extrinsic calibration methods, this paper introduces an online calibration algorithm for RGB-T cameras. This algorithm enables the extrinsic calibration of RGB-T cameras using only a pair of images from the same scene. To resolve the cross-modal feature matching problem, phase consistency is utilized in conjunction with MIM for feature detection and matching. Comprehensive experimental results demonstrate that the proposed intrinsic calibration method facilitates clear imaging for both RGB and thermal modalities, meeting the intrinsic calibration needs of RGB-T cameras. Additionally, the proposed online extrinsic calibration algorithm streamlines the calibration process, eliminating the reliance on calibration boards. Moreover, the calibration accuracy of our algorithm is comparable to that of offline calibration methods based on calibration boards. This means that in scenarios where external parameters are unstable or require online updates, our algorithm offers a significant advantage over traditional methods.

This study holds significant importance for applications in fields of computer and electrical engineering, including autonomous driving, surveillance, power inspection, robotics, and more. Our algorithm uses scene feature points for extrinsic calibration; therefore, in scenes with fewer feature points, the accuracy of the algorithm may decrease, potentially affecting the execution of subsequent tasks. To address this issue, future work could explore more robust feature extraction and matching methods, or enhance the calibration accuracy in low-texture scenarios by integrating additional information such as scene geometry or motion models. Besides, although this calibration method performs well in controlled environments, it may face certain challenges in real-world applications such as autonomous driving. These challenges mainly arise from lighting variations, dynamic scene changes, and sensor noise, which could reduce the effectiveness of cross-modal feature matching and impact calibration accuracy. To address these challenges, future research could focus on enhancing the algorithm's robustness to lighting changes, improving its adaptability to dynamic scenes, and incorporating techniques to handle sensor noise for further optimization.

CRedit authorship contribution statement

Jiahui Wei: Conceptualization, Methodology, Software. **Zhongwen Zou:** Investigation, Formal analysis. **Wenjie Lai:** Validation, Writing. **Junliang Du:** Review & Editing. **Jiaxue Zhao:** Resources, Data curation. **Ziji Liu:** Project administration, Funding acquisition. **Shengzhe Wang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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